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Abstract

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1

Introduction

Modern vehicles generate large amounts of telemetry data through distributed embedded systems, sensors, actuators, and in-vehicle networks [1]–[3]. These systems are designed to support real-time functionality, but they are also constrained by limited communication bandwidth, storage capacity, and on-board compute resources [2]. In practice, this has led to the widespread use of event-triggered and threshold-based logging strategies, which reduce data volume by transmitting only selected events or diagnostic messages [2], [4]. While these approaches are effective for limiting network load, they also risk omitting information that may later be relevant for offline analysis or machine learning-based applications [2], [5].

This challenge has become more important as vehicles generate increasingly rich signal-based data and as downstream machine learning tasks have become a central use of automotive telemetry [6]–[8]. Collected vehicle data is no longer used only for diagnostics or monitoring, but also for tasks such as predictive maintenance, anomaly detection, fleet analytics, and other forms of offline model training [8]–[10]. For these use cases, data utility matters as much as data volume: aggressive reduction of telemetry may improve efficiency, but it can also degrade the performance of models trained on the resulting data [11]–[13]. This creates a fundamental tension between efficient storage and transmission on the one hand and preserving task-relevant information on the other.

The theoretical basis for this tension comes from information theory and rate-distortion theory [14], [15]. Information entropy describes how much uncertainty is contained in a signal and therefore provides an indication of how compressible that signal is [14]. Rate-distortion theory extends this idea to lossy compression by formalizing the trade-off between bitrate and reconstruction quality [15]. For machine learning settings, this trade-off is not sufficient on its own, because good reconstruction does not necessarily imply that the compressed representation is useful for downstream tasks [16], [17]. This is why modern learned compression methods are often framed in terms of autoencoders and information bottlenecks: they aim to produce compact latent representations while retaining the information that is relevant for later prediction or classification tasks [18]–[21].

Within this thesis, neural compression is the central approach for addressing this problem. Neural compression methods use learned encoder-decoder architectures to transform high-dimensional input signals into more compact representations. In contrast to traditional algorithmic compression methods, these models can adapt to

the structure of the data and can be optimized for the kinds of signals that occur in automotive telemetry [18], [22], [23]. In particular, vector quantization offers a practical way to introduce discrete bottlenecks that reduce entropy while still allowing the model to preserve useful structure in the compressed representation [18], [24]. This makes neural compression especially attractive for edge-oriented settings, where the compressed data must be stored, transmitted, or processed under strict efficiency constraints [24], [25].

Recent research shows that neural compression has been successful in a range of domains, including time-series, audio, speech, and other resource-constrained applications [22]–[29]. For time-series data, continuous-latent autoencoder approaches have demonstrated strong reconstruction performance, while vector-quantized methods such as EdgeCodec show that lightweight discrete compression can be effective in edge-deployment scenarios [22], [24]. At the same time, the literature also shows that many methods primarily optimize reconstruction error, whereas downstream task utility is often treated as secondary or is not investigated in depth [22], [24]. For vehicle telemetry, this is a crucial limitation, because a representation that reconstructs well is not necessarily the best representation for predictive maintenance or anomaly detection [8], [9], [11].

The background of this thesis therefore combines three strands of work: the properties and constraints of automotive telemetry, the information-theoretic foundations of compression, and the recent development of neural compression methods. Taken together, these areas motivate the central question of the thesis, namely how to compress multimodal vehicle telemetry in a way that remains useful for downstream machine learning tasks while still achieving meaningful compression and efficiency gains. The following sections build on this background by stating the concrete aim of the thesis, defining the research questions, and outlining the limitations of the study.

1.1 Aim

The aim of this thesis is threefold. First, two existing neural compression methods are applied to automotive and IoT time-series data in order to evaluate how they perform in terms of compression ratio, reconstruction quality, and computational efficiency. One of these methods is based on vector quantization, while the other serves as a continuous-latent baseline.

Second, the compressed representations produced by these methods are used to train downstream machine learning models and are compared against models trained on uncompressed data. This makes it possible to assess how much task-relevant information is retained after compression and reconstruction.

Third, the two methods are further explored through changes to the quantization strategy and loss function. The goal of this exploratory part is to investigate whether reconstruction loss and downstream utility retention can be improved further without sacrificing the general compression objective and computational efficiency.

1.2 Research Questions

This thesis is guided by the following research questions:

1. RQ1: How do the selected neural compression methods perform on automotive and IoT time-series data in terms of compression ratio, reconstruction quality, and computational efficiency?
2. RQ2: To what extent do downstream machine learning models trained on reconstructed compressed data retain task performance compared to models trained on uncompressed data?
3. RQ3: Where do the existing neural compression methods fall short in terms of reconstruction quality and downstream utility retention, and which quantization strategies and loss functions can mitigate these limitations?

1.3 Limitations

Due to time constraints and the scope of this thesis, there are several limitations to the work presented.

First, the work forgoes testing on real hardware as that is very time consuming and difficult to set up while contributing relatively little to the main research question. Instead, canonical methods to measure computational efficiency are employed.

The work is further limited to testing only regression tasks and anomaly detection as downstream tasks due to time constraints and the structure of the available datasets. These two tasks should, however, be a good representation of common downstream tasks, and adding more tasks would likely not drastically change the conclusion of the thesis.

The last noteworthy limitation is that this work does not measure computational efficiency of the evaluated neural compression models. This choice is driven by two practical reasons. First, implementing and benchmarking on representative edge hardware is too complex and time consuming for the scope of this thesis. Second, EdgeCodec's prior work already demonstrates an efficient encoder design, which, importantly, we do not change in the work presented within this thesis. For these reasons, explicit runtime or hardware measurements are considered out of scope for this thesis.

2

Related Works

Within the neural compression research community, several distinct research directions are relevant to this thesis. This chapter outlines those directions, presents representative examples, and identifies the research gaps addressed by the present work.

Time-Series Neural Compression Neural compression for time-series data has been studied, but many methods are not optimized for edge deployment. The architecture proposed by Zheng and Zhang [22] is representative of this line of work. Their method uses a continuous-latent autoencoder with recurrent modules and is primarily optimized for reconstruction quality, while computational efficiency and downstream task utility retention are not central optimization targets [22].

Edge-Optimized Neural Compression The majority of research labeled as "edge-optimized neural compression" focuses on compressing ML models for deployment rather than compressing input data. Lai et al. [30] highlight the importance of enabling complex deep learning models on resource-constrained edge devices. Key developments include improved efficient transformer variants and knowledge-distillation methods [30]. These advances address one side of the edge-optimization problem (model compression), while the complementary side is efficient compression of the data itself. EdgeCodec is a notable example of this latter direction and is discussed below.

Audio/Speech Neural Compression The audio/speech community contains extensive research on neural compression, with a strong emphasis on vector quantization. A key development in this literature is residual vector quantization (RVQ). RVQ works by layering hierarchical quantizers whith each layer quantizing the residuals of the previous layer [27, pp. 495–507].

- SoundStream (Google, 2021): first major use of RVQ for audio compression [27, pp. 495–507].
- EnCodec (Meta, 2022): refined RVQ for high-quality audio [28].
- WaveTokenizer (2024): recent RVQ-based audio codec [29].

EdgeCodec EdgeCodec transfers RVQ ideas from audio/speech processing to time-series compression (aerospace sensor data) and is explicitly designed for edge deployment. This is achieved through a lightweight encoder deployed on the edge and a heavier decoder executed in the cloud [24]. As a result, EdgeCodec is a notable

outlier in neural compression literature and addresses many challenges apparent in the research field. However, it has important limitations for the present use case: it is optimized for smooth, continuous aerospace sensor streams rather than mutlimodal vehicle telemetry, and it prioritizes reconstruction error over downstream task utility.

The literature review indicates that neural compression, VQ-based quantization, and lightweight design principles have all been explored individually. The remaining gap is a method tailored to mutlimodal data that explicitly optimizes downstream ML task utility rather than reconstruction quality alone. Based on this gap, the present thesis evaluates two existing solutions, namely the continuous-latent autoencoder by Zheng and Zhang [22] and EdgeCodec [24], with respect to downstream ML task performance. Baseline models derived from these works are implemented and evaluated on automotive telemetry data, with the objective of identifying and improving design choices for multimodal data and downstream utility retention.

3

Theory

This section introduces necessary theory and establishes important context for further discussion within this thesis. The in-vehicle system and related components will be introduced, followed by an introduction to modern challenges in regard to data processing and data usage in the automotive industry. Afterwards the most important information theory concepts will be introduced, so that finally neural compression as an idea can be outlined with its most important methods and techniques.

3.1 In-Vehicle Systems and Event-Triggered Logging

The In-Vehicle Embedded System An embedded system is a computer system integrated into a physical device or product to perform a dedicated function as part of a larger system [1, pp. vii–x, 1–3]. The in-vehicle embedded system is a specialized embedded system integrated within a vehicle. It is essential for real-time functions like controlling, monitoring, and enhancing various automotive operations [2, pp. 1-1–1-5]. These in-vehicle embedded systems, like embedded systems in general, typically consist of both hardware and software components, such as electronic control units (ECUs), sensors, actuators, and communication interfaces [1, pp. 3–4], [2, pp. 1-3–1-10].

ECUs serve as the primary processing nodes in in-vehicle embedded systems. They are responsible for executing control algorithms on sensor inputs and issuing commands to actuators. They also handle local control logic while coordinating via in-vehicle networks to form a distributed system [2, pp. 1-8–1-19]. An example of how ECUs work together to create the in-vehicle embedded system is illustrated by [2, pp. 1-8–1-9]. In the example an ECU coordinates a request, e.g., locking or unlocking a door, while the controller realize the request on the physical system, which is controlled by another ECU. The resulting system is the main generator of telemetry data within the vehicle. Vehicle telemetry data, usually defined as data that is produced by the vehicle and sent off-board, is key to improving important aspects of these vehicles like safety [3, pp. 1–5]. This vehicle telemetry data is at the core of this thesis.

In-Vehicle Networks As mentioned above, in-vehicle networks serve as the connectors between the ECUs of the in-vehicle embedded system. Among these in-vehicle networks Control Area Networks (CAN), Local Interconnect Networks (LIN), FlexRay,

and Ethernet are popular representatives that enable efficient communication and coordination among different vehicle subsystems [2, pp. 1-2–1-5].

From a technical perspective, ECUs connect to these networks via dedicated communication services. The communication services are tasked with providing a uniform interface to the network, while hiding protocol and message properties from the application [31, p. 48]. The in-vehicle network protocols are categorized into three classes, A, B, and C. LIN falls into category A with a bit rate below 10 kbps and application in sensor and actuator networks. CAN-B is a representative of category B with a bit rate between 10–500 kbps. Category B networks serve vehicle-centric domains and body electronics. Category C in-vehicle networks like FlexRay or CAN-C are defined by having a bit rate of lower than 1 Mbps and used in safety-relevant systems [2, pp. 1-12–1-13]. Despite their effectiveness in their respective use-cases, all in-vehicle networks must contend with fundamental bandwidth limitations that constrain data throughput across the vehicle [2, pp. 1-18–1-19].

Event-Triggered and Time-Triggered Logging Event-triggered logging and diagnostic frameworks, which record data only when specific error codes occur or threshold crossings are detected, are widely adopted to reduce data transmission and avoid bus saturation in complex systems such as the in-vehicle embedded system [4, pp. 7–13]. According to the AUTOSAR Diagnostic Log and Trace (DLT) standard, this mechanism can be implemented as follows: When software components or Basic Software modules report errors via the Default Error Tracer, the DLT module receives these via a specified function and stores timestamped log/trace messages to flash storage [4, p. 25]. Non-critical events are filtered during normal operation [4, pp. 30–35].

Time-triggered logging serves as an alternative to event-triggered logging and works by setting predetermined points in time where data is transmitted. The validity of this strategy is easily monitored, as the time-frame scheduling is predictable and missing messages are immediately identified [2, pp. 4-5]. Although Navet discusses these paradigms in the context of communication scheduling, similar principles apply to diagnostic event- and time-based mechanisms, which leads to similar struggles in these systems.

While these approaches have been effective in reducing the data load of the in-vehicle networks, they come with major limitations. Two immediate issues these diagnostic frameworks struggle with is presenting an unbiased and comprehensive monitoring of the system, while having to carefully tune event thresholds and diagnostic parameters [2, pp. 4-5–4-6], [5]. In the next section we will go further into how these limitations are accentuated by recent developments.

3.2 Downstream Machine Learning Tasks and Data Quantity

Two developments in recent years further underline the shortcomings of event-triggered logging in automotive systems: the massive increase in signal-based data

in the in-vehicle network and the growing relevance of downstream machine learning (ML) tasks.

Data Quantity in Modern Vehicles Recent industry and research reports indicate that the data quantity generated by sensors and the amount of interconnectivity in vehicles is growing at an extremely rapid pace. According to a 2021 automotive electronics report, by 2030, about 95% of new vehicles will be connected, up from around 50% today [6]. Another report states that even today, a single car can generate up to 1 terabyte (TB) of data per hour from its sensors [7]. This explosive growth is driven by the increasing number and sophistication of sensors—such as cameras, radars, and lidars—required for advanced safety and autonomous driving features, with the complexity and volume of data presenting significant challenges for storage, processing, and transmission within embedded automotive systems [7], [32, pp. 3–5]. While much of the focus in research and reporting on this issue centers on autonomous driving technologies like lidars and cameras, it is important to note that the core of the in-vehicle system, the ECU, is also a major contributor to the data quantity issue. This implies that not only autonomous driving technologies but modern vehicles in general are facing these challenges [32, pp. 3–4].

Downstream ML Tasks Modern vehicles increasingly rely on data-driven intelligence to enhance safety, reliability, and efficiency. As there currently exist no conclusive definition of these tasks, we will refer to them as downstream ML tasks within the context of this thesis. These downstream ML tasks we define as the offline use of collected vehicle data for the training of ML models for various tasks. This is distinctly different from real-time perception and control systems. Downstream ML tasks have become central to automotive-system design. Theissler et al. [8, p. 2] mention three reasons for the rise of importance of these downstream ML tasks: data availability, breakthroughs in algorithm development and the advancements in computational power. Prominent examples of downstream ML tasks in the automotive context include: Predictive maintenance tasks aim to predict the expected time of a failure, thereby helping to estimate remaining functionality of components or systems [8, p. 3]. Anomaly and intrusion detection tasks want to find anomalies or outliers in data points which is especially relevant in fragile systems like the CAN network [9, pp. 1–3]. Fleet-level management operations like fuel consumption that aim to better utilize important assets in vehicle systems [10, pp. 1–2]. Utility loss in the datasets used to train the models for these tasks translates directly to degraded model performance, increased operational costs, and missed business opportunities [32, pp. 4–5]. This further underlines the importance of a comprehensive data logging strategy.

There exists relatively little research on the use of event-triggered logging systems as a compression strategy and it's impact on downstream ML tasks. It is therefore difficult to definitively say that event-triggered logging systems specifically produce data with lower utility. Studies have however shown that similar utility agnostic data reduction strategies can have negative impacts on downstream utility in the domain of time-series data. One such example comes from a study by [11] where the authors show that a 10-minute aggregation of the data leads to degraded utility for downstream ML tasks. In this specific case, the F1-score of the predictive

downstream model was reduced by from 0.63 to 0.50 when trained on the aggregated data compared to the original dataset. The same study also shows that a model based lossy compression technique can be implemented without significant utility loss which suggests that more strategic data reduction techniques can retain more utility for downstream ML tasks. Further analysis of the impact of lossy compression algorithms on time series task utility in general has been done by [12] and [13]. It is important to note, that, while challenging, at specific error bounds compression techniques within these studies can actually show improvement to the task metrics. All this highlights the need for a strategic lossy compression technique that accounts not only for reconstruction, but also for downstream task utility.

3.3 Compression Theory and Traditional Methods

Information Theory Because information theory provides the basis for compression methods, its core concepts are introduced here before discussing compression itself. Entropy is key to information theory. It measures the uncertainty of a random variable and is computed from its probability distribution. More specifically, the uncertainty in a probability distribution is the expected negative log-probability of its outcomes. The entropy of a discrete distribution $p = (p_i)_i$ is defined as

$$H(p) = -\sum_i p_i \log p_i,$$

where the base of the logarithm determines the unit [14].

To illustrate with a weather example, suppose the true probabilities of "rain" and "shine" are $p_1 = 0.3$ and $p_2 = 0.7$, respectively. Then (using base-2 logs),

$$H(p) = -(p_1 \log p_1 + p_2 \log p_2) \approx 0.61.$$

Suppose the measurements are instead from Abu Dhabi, where the probabilities might be $p_1 = 0.01$ and $p_2 = 0.99$. The entropy is now approximately 0.06. There is thus much less uncertainty about any given day compared to a place where it rains 30% of the time. In this way, information entropy quantifies the inherent uncertainty in a distribution of events.

Entropy is central to compression because it indicates how compressible data is. High-entropy data is less predictable and therefore harder to compress, while low-entropy data has redundant patterns and is therefore easier to compress [14].

The other important information theory concept to understand in the context of compression is mutual information. Mutual information, formally defined as $I(X;Y)$ answers the question "How much does one variable tell me about another?" [14].

Compression Compression, as originated in information theory by [14], is the process of encoding information using fewer bits than the original representation. Compression techniques can be broadly categorized into lossless and lossy methods. Lossless compression is based on two principles: distribution modeling, sometimes called entropy modeling, and entropy coding. Entropy modeling involves creating

a probabilistic representation of the data, while entropy coding assigns shorter codes to more frequent symbols based on their probabilities, thereby minimizing the average code length. Entropy serves as the lower bound (in bits per symbol) for any lossless compression scheme. Lossy compression allows for some loss of information in exchange for higher compression ratios. This is typically achieved through techniques such as transform coding and quantization [33]. Closely connected to lossy compression techniques is the term "compression ratio". Compression ratio is defined as the size of uncompressed data divided by the size of compressed data [34]. For the purpose of this thesis, the focus will be on lossy compression as this is more suitable for common downstream ML tasks where some loss of fidelity is acceptable as long as the relevant information for the task is preserved.

Rate-Distortion Theory Rate-distortion theory provides the fundamental framework for lossy compression, quantifying the minimum bitrate $R(D)$ required to represent data with acceptable distortion at most D. Building on entropy and mutual information introduced earlier, it formalizes the fundamental tradeoff between rate and fidelity central to lossy compression. The rate-distortion function is formally defined as

$$R(D) = \min_{p(\hat{X}|X):\ E[d(X,\hat{X})]\leq D} I(X;\hat{X}),$$

where $d(x,\hat{x})$ is a distortion measure, e.g., squared error, $E[\cdot]$ is the average over all data points or expectation, and the minimum bitrate is computed over all distributions $p(\hat{X}|X)$ achieving average distortion at most D. Shannon's rate-distortion theorem proves this isn't just a theoretical lower bound and practical codes exist that get arbitrarily close to $R(D)$ bits/symbols while keeping distortion below D. [15].

Distortion measures quantify the cost of representation between original data X and its compressed reconstruction $\hat{X}$. They are essential in rate-distortion theory, where the goal is to minimize bitrate subject to $E[d(X,\hat{X})] \leq D$ for a chosen distortion threshold D. Widely used examples for distortion measures include the squared error (MSE) or the Hamming distance. MSE is defined as $d(x,\hat{x}) = (x-\hat{x})^2$, and is often used for numerical/time-series data due to its mathematical tractability. The Hamming distance on the other hand is defined as $d(x,\hat{x}) = \sum_i \mathbf{1}\{x_i \neq \hat{x}_i\}$, and is used for binary or categorical data [35, pp. 304–305].

In the ML context, particularly in automotive systems, or in fact Internet-of-Things (IoT) systems in general, rate-distortion analysis denotes the trade-off between efficiently handling large quantities of data and maximizing model performance. So in this sense, rate-distortion theory gives the theoretical framework to the downstream ML task challenges described in Section 3.2. Traditional lossy compression techniques can reduce data volume, but often at the cost of losing critical information necessary for accurate ML tasks such as predictive maintenance, anomaly detection, and fleet analytics [16]. The impact of this trade-off is well-documented in the literature. Cuza et al. [17] for example, study the impact of lossy compression techniques on time series forecasting tasks and observe a constant trade-off between compression ratio and forecasting accuracy. It is important to note, that, while the authors of this paper see impact of compression ratio on task performance, it is not necessarily a negative impact and is dependent on various factors [17, pp. 654–655].

Traditional Compression Methods Traditional compression methods, based on information theory principles, often fall short in automotive applications, especially as a precursor for downstream ML tasks. For video/image compression, traditional methods like JPEG are optimized for human perception (e.g., visual quality) rather than ML tasks or efficient downstream data use [36, p. 12]. For time series data, algorithmic approaches like CHIMP or Gorilla depend on manually chosen parameters like window size and are sensitive to data characteristics such as entropy and signal variability. This limits their effectiveness in capturing the nuances required for accurate ML model performance in automotive contexts [37, pp. 37–38 and 47]. These algorithmic approaches were investigated by Johnsson [37] in a previous Master's thesis project. This work builds upon this thesis by exploring neural alternatives for vehicle telemetry compression. These neural alternatives are more flexible and can be optimized for specific downstream tasks and data characteristics, potentially offering better performance in terms of the rate-distortion trade-off for ML tasks. As previously shown by [11], model aware compression techniques have the potential to retain more task relevant information compared to traditional methods, which is key for achieving better performance for downstream ML tasks while still maintaining significant compression ratios.

3.4 Neural Compression

In the most basic sense, neural, or sometimes learned compression, is the application of neural networks and related machine learning techniques to the task of data compression [18]. Neural compression has been shown to be an effective strategy for handling the data rate-utility trade-off in various domains. Studies as early as 2019 have shown that neural compression methods can outperform traditional compression techniques for image and video data, especially at low bitrates [26]. The same has been shown for time series data [22], [23]. This approach has also been shown to be effective in resource-constrained environments, such as IoT systems, where the ability to retain task-relevant information while reducing data volume is crucial [24], [25].

In the following sections, the basics of neural compression theory and the connection to relevant information theory concepts are outlined.

Information Bottleneck Theory While the theory presented by [14] gives a theoretical framework for understanding the limits of compression, it falls short when it comes to learned compression as it only considers the maximum compression ratio for a given distortion level, without considering the specific characteristics of the data or the downstream tasks. This shortcoming is addressed in the information bottleneck (IB) theory, which presents a more nuanced approach to the issue of compression. In order to maintain relevance for downstream tasks, we need to preserve as much relevant information as possible about the target label, not about the input data. This is the core idea of the IB theory, which formalizes the trade-off between compression and relevance as:

$$\min_{p(\hat{X}|X)} I(X;\hat{X}) - \beta I(Y;\hat{X}),$$

where $I(X;\hat{X})$ is the mutual information between the input data X and its compressed representation $\hat{X}$, $I(Y;\hat{X})$ is the mutual information between the target label Y and the compressed representation $\hat{X}$, and β is a Lagrange multiplier that controls the trade-off between compression and relevance [19].

Early work on attempting to understand the learning dynamics of deep neural networks (DNNs) have applied the IB theory in an attempt to explain how DNNs learn compressed representations of data while preserving relevant information for specific tasks [38, pp. 1-5]. The core of this idea is that the goal of any supervised learning algorithm is to capture a minimally sufficient statistic in the input required to express the output. This mirrors the goal of IB compression which claims that there exists a maximally compressed representation of the input which retains the maximum amount of information about the target label. Other works such as [39] have attempted to show that DNNs both fit the input in order to increase the mutual information between the target and the latent representation $I(T;Y)$ while also compressing the input by lowering the mutual information between the input and the latent representation $I(X;T)$. This work has however been criticized for its methodology and computation of mutual information. Much of the issue comes down to the fact that the layers of a DNN cannot lower mutual information, but rather just learn to map the input to consistent output values, effectively decorrelating the data structure. This process will be referred to as geometric compression. In order for a network to be able to discard information and thereby compress the input in an IB sense, additional measures are required [40], [41]. So how DNNs learn is debated. But it becomes clear that geometric compression, while being dependent on the architecture or the used activation functions, is key to how DNNs function. Modern neural compression research therefore almost always combines geometric compression supplemented by some form of information-theoretic compression.

Autoencoders & Vector Quantization The implementation of lossy neural compression methods is firmly based on autoencoders (AEs) and more specifically on Rate-Distortion Variational Autoencoders or R-D VAEs [18, pp. 145–146]. Generally AEs are defined as DNNs that try to reconstruct their input. This implies that the target output is the input. For this, AEs are composed of an encoder, which produces a hidden or latent representation, and a decoder, which takes this representation as an input and tries to reconstruct the original input. AEs are trained end-to-end, which means that the encoder and the decoder are trained in tandem. The training goal is to make the latent representation take on meaningful properties [20].

The main difference between AEs and VAEs, introduced by Kingma and Welling [42], is that VAEs pick from a distribution of points when assigning latent representations, while AEs are deterministic in nature [20]. Importantly, many modern learned compression methods are formulated in a VAE-style rate-distortion framework. This is because an AE is essentially a VAE with a degenerate (Delta) variational posterior distribution. A deterministic autoencoder maps x to a single point $\hat{z} = f(x)$; the variational posterior $q(z|x)$ then becomes a Delta distribution and therefore deterministic [18, pp. 145–146]. The choice between using a stochastic encoder or a deterministic encoder with quantization is an ongoing debate in the research community. Because of the non-differentiability of quantization, and the resulting problems

in end-to-end training of lossy compression models, there has been experimentation with stochastic encoders. Theis and Agustsson [43] summarize the advantages of stochastic encoders, and while there certainly are advantages to stochastic encoders, deterministic encoders are favored by most modern neural compression methods [18], [43]. No matter which of these approaches are implemented, they both display the compromise between geometric compression and information-theoretic compression outlined above. In the case of deterministic encoders, which are the focus of this thesis, the information-theoretic compression is achieved via quantization, which will be introduced in more detail in the next paragraph.

Quantization is often used in neural compression to introduce true information-theoretic compression when using deterministic encoders. While there have been various approaches to achieving true compression of information in the context of IB using neural networks, such as the variational information bottleneck (VIB) architecture [21], the approach that is presented in this thesis extends the traditional autoencoder architecture with a VQ bottleneck in order to minimize the entropy. This idea is conceptually more similar to methods such as those presented in [44, pp. 1611–1630]. The VQ layer acts as a deterministic information bottleneck partitioning the input space X into regions that map to the same discrete code t, discarding intra-cluster variations. By binning the continuous signal to a finite set of codes, the entropy of the signal is bounded by the number of bins. As mentioned in [40], the encoder cannot exhibit decreasing $I(X;T)$ without noise or stochasticity since the entropy is constant $H(T \mid X)$. It can however create sharper decision boundaries which allow the VQ bottleneck to more effectively quantize the input without discarding relevant information.

In neural compression methods, autoencoder architectures are typically chosen based on the structure of the input signals, the computational restraints, and the target rate-distortion objective. For telemetry and other mutlimodal time series data convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are popular design choices [18, pp.146–148]. CNNs naturally excel at capturing local temporal patterns efficiently, while RNNs are used to model longer-range dependencies [18, pp.146–148]. Both CNNs and RNNs are especially suitable for neural compression, because they use a more compact parametric representation than for example fully connected neural networks [45]. Naturally, a lot of hybrid systems trying to combine CNNs and RNNs have also emerged over the years. In practice, this architectural choice is not only a modeling decision but also a systems decision. Lightweight designs usually avoid costly recurrent designs and prefer convolutional approaches when compression is deployed on resource-constrained edge devices with strict latency and memory budgets [24], [46].

4

Methods

This chapter describes the methodology used in this thesis. It outlines the architectures that are analyzed, how they are implemented, and which improvements are made to align them with the stated goals. It also presents the empirical research strategy, and experimental setup used to evaluate these approaches.

4.1 Experimental Strategy

To answer the research questions outlined in Chapter 1, this thesis follows a three-phase experimental strategy. Following the taxonomy introduced by Stol and Fitzgerald [47], this thesis uses a controlled laboratory experiment to compare neural compression architectures under a fixed benchmark setup. The emphasis is on isolating the effect of architectural choices on the defined metrics.

First, two baseline neural compression methods are selected to represent the main design directions discussed in the related work, namely a continuous-latent method and a discrete-latent method. Zheng and Zhang [22] are used as the representative continuous-latent baseline, while EdgeCodec by Hodo et al. [24] is used as the representative discrete-latent baseline because of its emphasis on efficient edge-oriented compression.

In the second phase, both baseline methods are reproduced according to the specifications of the respective authors and evaluated on IoT and automotive time-series data. Their compressed representations are then used to train downstream machine learning models, which are also trained on the corresponding uncompressed data. Comparing the downstream performance of these models provides a direct measure of how much utility is retained after compression and reconstruction.

In the third phase, the two baseline methods are examined in greater detail to identify their main failure modes with respect to reconstruction quality and downstream utility retention. Based on these findings, targeted modifications to the quantization strategy and loss function are implemented in two new exploratory variants, named *UniEdgeCodec* and *TAEdgeCodec*. These variants are then evaluated using the same experimental setup as the original baselines in order to assess whether the proposed changes improve the observed limitations.

4.2 Threats to Validity

The validity of empirical software engineering research is commonly divided into internal validity, construct validity, conclusion validity, and external validity. Internal validity concerns whether the observed effect is truly caused by the treatment or whether it can be explained by other uncontrolled factors. In controlled laboratory experiments, such as the experiments conducted in this thesis, this is a central concern. Although confounding factors can still occur, the experimental setup is designed to minimize them and isolate the causal effect.

External validity usually suffers when the focus is on internal validity. External validity concerns the generalizability of the findings. While different datasets and use cases are used to provide some variation, the focus of this thesis is to observe the effects of architectural choices on selected metrics. The external validity is therefore limited and should be examined more carefully in future work.

Conclusion validity, which concerns the soundness of the empirical findings, is an important issue in experiments that depend on many hyperparameters and training parameters. To avoid fishing for satisfactory results or p-hacking, the experimental setup is fixed as much as possible before the experiments are run. This helps to preserve conclusion validity, but the results may still differ when the setup is changed.

Lastly, construct validity, or conceptual correctness, concerns whether the chosen metrics actually measure the intended concepts. In this thesis, compression quality is operationalized through reconstruction error and compression ratio, while utility retention is approximated through downstream task performance after reconstruction. These measures are standard in the neural compression literature, but they only capture the intended constructs indirectly. For example, low reconstruction error does not necessarily imply semantic preservation, and downstream performance depends not only on the compressed representation but also on the decoder and the selected downstream tasks.

4.3 Comparison and Metrics

There are two dimensions along which the compression methods are compared. First, the *compression performance* of both methods is assessed. The metrics compression ratio (CR) and reconstruction error are primarily used to evaluate compression performance. These metrics align with the evaluation standards in the neural compression literature. Additioanlly, average correlation between the reconstructed signals and perceptual reconstruction quality are taken into account, to give a broader picture of the qualities of the reconstructed artifacts. All metrics are measured by training the models on a reconstruction task and evaluating the reconstruction quality and effective CR. For reconstruction error, mean squared error (MSE) is used. Within this thesis, the mean-reduced variant of MSE is used. The MSE loss is defined as:

$$L_{\text{MSE}} = \frac{1}{T \cdot D} \sum_{t=1}^{T} \sum_{d=1}^{D} \left(x_t^d - \hat{x}_t^d \right)^2 \tag{4.1}$$

Besides reconstruction error, CR is the other important metric to evaluate compression performance. Within this thesis, CR is defined exactly as described by the authors of the EdgeCodec paper [24]. CR is defined as:

$$\#\text{bits} = \lceil \log_2(k) \rceil, \quad \text{CR} = \frac{A \cdot B}{\#\text{bits} \cdot N} \tag{4.2}$$

Because this equation is not immediately intuitive, a numerical example is provided. The architecture presented by Hodo et al. [24] compresses the input using both the encoder and the quantization module. In the EdgeCodec paper, the input has 36 channels and a window size of 800 time steps. The output of the encoder is a downsampled version of this input with dimensions 9 channels $\times$ 100 time steps. This means that the encoder alone compresses with a CR of $(800/100) \cdot (36/9) = 32\times$. In the quantization setup, the authors compress with precision A, which corresponds to 32-bit precision of the input signals. The number of bits per code depends on the codebook size k. For a codebook of size 768, the bits per code are roughly 10. The total number of transmitted codes, N, is calculated by multiplying the number of quantizers by the number of input channels to the quantizers. Because the encoder outputs 9 channels and the architecture presented by Hodo et al. [24] uses four quantizers, N is equal to 36. Applying the CR equation gives $\frac{9 \cdot 100 \cdot 32}{36 \cdot 10} = 80\times$. The total CR is then obtained by multiplying the encoder CR by the quantizer CR, resulting in a total CR of $32 \cdot 80 = 2560\times$, as presented in the paper.

Second, the *downstream task performance* when using compressed representations of the training data for selected downstream models from both methods is evaluated. This includes metrics such as accuracy for classification tasks or MSE for regression tasks. To measure this, the downstream models are trained on uncompressed data and compressed data using different neural compression methods. By comparing the utility loss across these approaches, the utility retention of the various neural compression methods can be measured.

An important note regarding comparability of the neural compression methods, especially for downstream task performance, is that the methods produce different types of compressed representations. The TCN-RNN baseline produces a continuous latent representation, while the EdgeCodec implementation produces discrete latents. To ensure a fair comparison, the compressed data is reconstructed using the learned decoders before being used as training data for the downstream models. This setup makes it possible to measure which approach retains more downstream task utility after compression and reconstruction.

Another important note for the evaluation of downstream task performance is that only the test portion of the available datasets is used when training downstream models. The train and validation splits are used during training of the neural compression methods. The test set is then used to evaluate the neural compression method.

Because the neural compression methods have seen the train and validation splits during training, these splits cannot be used when deploying the neural compression methods to compress data for downstream tasks. For this reason, only the test split is used as input, either compressed or uncompressed, to the downstream models.

4.4 Data

The goal is to use industry-relevant data while also ensuring that the methods are tested on publicly available data. For this reason, two datasets are used: the Volvo electric mobility (EMOB) dataset and the appliances energy dataset used by Zheng and Zhang [22], [48]. Both datasets are accessed through .csv files. The Volvo EMOB dataset captures battery behaviour within an embedded truck system. Three key signals are captured: "Temp 1" for battery temperature and "Amps" and "Volts" for battery current and voltage. Figure 4.1 shows an excerpt of the test dataset, demonstrating the characteristics of each signal used in the Volvo EMOB dataset. The appliances energy dataset measures appliance energy use in a low-energy building over 4.5 months. Key signals used in this thesis are "Appliances," which measures energy use in Wh and is the primary regression target of the dataset; "lights," which measures the energy use of light fixtures in the house in Wh; "Press_mm_hg," which measures outside pressure in mm Hg; "T_out," which measures outside temperature in Celsius; "RH_out," which measures outside humidity in percent; "Windspeed," which measures outside wind speed in m/s; "Visibility," which measures outside visibility in km; and "Tdewpoint," which measures outside dewpoint temperature. Figure 4.2 presents a snippet of the test split of the appliances energy dataset, showing the behavior of each used signal.

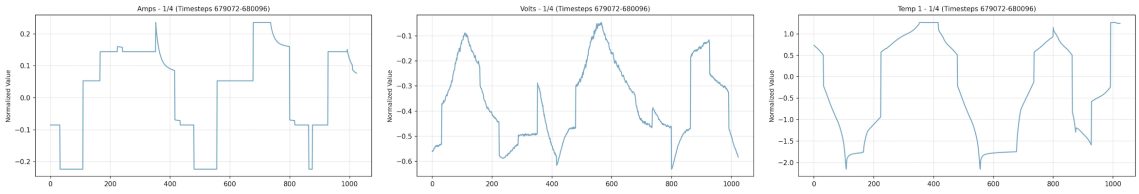
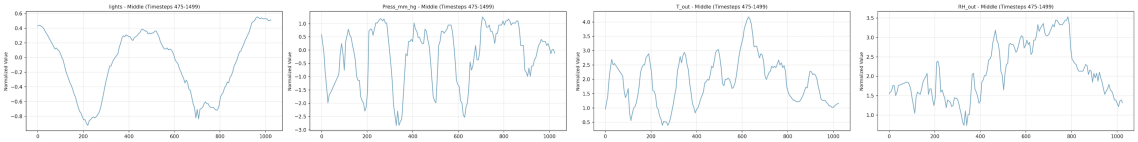


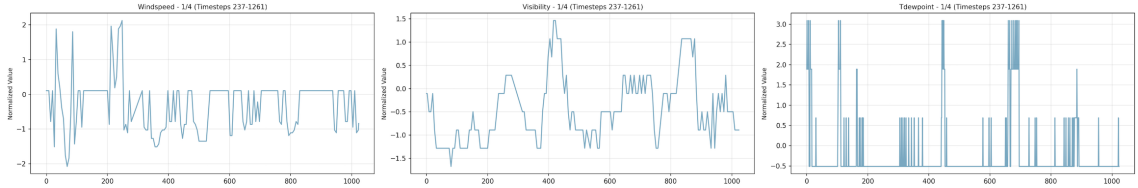
Figure 4.1: Visualization of the Volvo EMOB Dataset

For input handling and preprocessing, it is important that the time-series data is segmented into fixed-length windows. These windows are treated as inputs to the model. Overlapping windows can be used to increase the effective training data size and improve reconstruction quality at window boundaries. The appliances energy dataset uses a window size of 144 with stride 1, and the Volvo EMOB dataset uses a window size of 128 with stride 64. Additionally, each channel is normalized independently using statistics computed on the training set. The train/val/test split for the Volvo EMOB and appliances energy datasets is 0.4/0.1/0.5 and 0.81/0.09/0.1, respectively. The values for window size and split ratio for the appliances energy dataset are taken directly from Zheng and Zhang [22]. For the Volvo EMOB dataset, these values result from experiments with different configurations and selection of suitable parameters. The split ratio for the Volvo EMOB dataset heavily emphasizes

the test set because, as outlined in the section on downstream task performance, only the test set is used to train downstream models. As the downstream use case for the Volvo EMOB dataset is non-trivial, a larger amount of data is needed for training.



(a) First four Signals of the Appliances Energy Dataset



(b) Last three Signals of the Appliances Energy Dataset

Figure 4.2: Visualization of the Appliances Energy Dataset

With the setup outlined in the previous paragraph, the two datasets have the following characteristics. The Volvo EMOB dataset is divided across a total of 50 files with 6189262 rows across these files. With a window size of 128 and a stride of 64, a total of 84911 windows are produced. Of these windows, 76993 are normal windows and 7918 are anomaly windows. The train set includes 33964 total windows, 30797 normal and 3167 anomaly windows. The validation set includes 8490 windows, 7699 normal and 791 anomaly windows. Lastly, the test set includes 42457 windows, 38497 normal and 3960 anomaly windows. There are no NULL values in the Volvo EMOB dataset. The appliances energy dataset has a total of 19735 rows. Because of the aggressive stride of 1, this results in 19592 total windows. Of these, 15985 are assigned to the train split, 1776 are assigned to the validation split, and 1974 are assigned to the test split. The appliances energy dataset also has no NULL values.

Each dataset is implemented using a PyTorch “Dataset” and a PyTorch Lightning “LightningDataModule”. Between these two components, the PyTorch “Dataset” is responsible for data preparation steps such as connecting to the data source, loading the data, filtering the data, computing global statistics, and creating windows. The PyTorch Lightning “LightningDataModule” encapsulates this logic and splits the data to produce machine-learning-ready train/validation/test datasets.

4.5 Downstream Use Cases

Comparing the compression methods based on downstream task performance requires concrete downstream use cases. The use cases and the models that implement them must be simple enough to enable clear evaluation of the compression methods, while still being relevant to real industry problems. For this reason, two downstream use cases are defined.

For the appliances energy dataset, the dataset-defined regression task of predicting the "Appliances" signal is implemented. All aforementioned signals are used as features, and reconstructed windows (decoder outputs) are fed to a random forest regressor. The random forest provides a balance between simplicity for comparison and adequate predictive power. The regressor is implemented with `scikit-learn` and initialized with `n_estimators=100` and `max_depth=10`. The regressor is evaluated using standard metrics such as MSE and R2 but the final utility retention comparison is based on the R2 metric because of its robustness and scale invariance.

The Volvo EMOB dataset is used for an anomaly-detection classification task. Labels are produced using a Volvo internal interquartile-range outlier detection script. This script uses the 3-sigma rule as a threshold to capture roughly 99.7% of normal data and classify the rest as anomalies. All three signals are used as features to train a random forest classifier, implemented identically to the regressor (same hyperparameters: `n_estimators=100`, `max_depth=10`). Because there is class imbalance in the Volvo EMOB dataset, the `class_weight='balanced'` parameter is additionally used. The classifier is evaluated using standard metrics such as accuracy, precision, recall, and F1-score. The final utility retention calculations for this use case will be based on the F1-score, as it provides a good balance between precision and recall, which gives a comprehensive picture of the model's performance on the imbalanced dataset.

4.6 Implementation Details

4.6.1 Baseline Implementations

In this section, the architectures of the established compression methods which are used as baselines are introduced in detail. If not stated otherwise in this section, all configurations for the models are identical to those of the respective authors. A detailed config file for each specific experiment is provided in the next chapter.

4.6.1.1 Baseline 1: TCN-RNN

The continuous-latent autoencoder approach introduced by Zheng and Zhang [22] serves as the basis for the continuous-latent baseline model. Their work includes two models, the TCN-RNN model and the TCN-ARNN model, as well as a model selection mechanism. For simplicity, the focus here is on the TCN-RNN model. The TCN-RNN model is representative of state-of-the-art continuous-latent neural time-series compression methods. Compared to vanilla autoencoders, its architecture is explicitly designed to capture long temporal dependencies and has demonstrated stronger rate-distortion performance. This makes it a suitable baseline for evaluating the proposed implementation against a competitive neural compressor. Another point to note about the TCN-RNN model is its costly architectural design, namely deep temporal convolutional stacks with large receptive fields, recurrent (often sequential) decoding, and dense continuous latent representations. These components typically lead to increased FLOPs, higher activation memory, and longer inference latency.

The TCN-RNN model is composed of four main components: the TCN-RNN encoder,

a linear projection that serves as the bottleneck, the GRU decoder, and a second linear projection. A complete overview of the TCN-RNN architecture is shown in Figure 4.3.

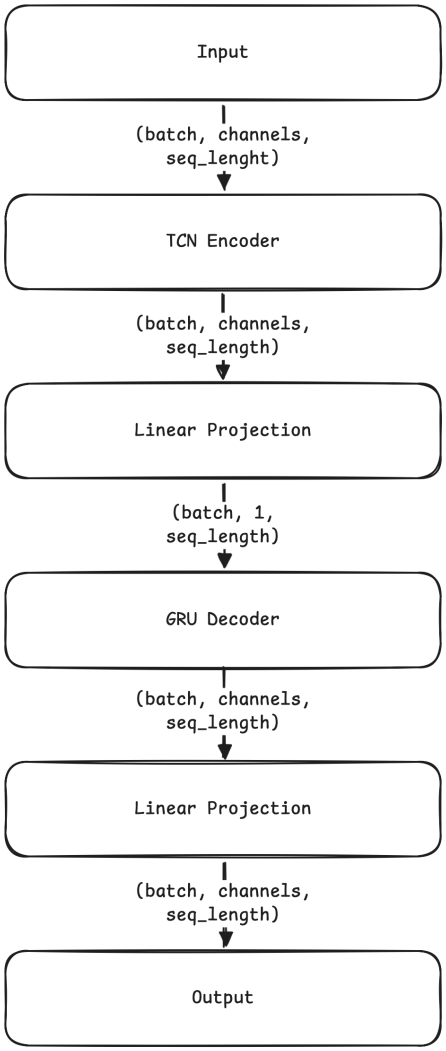
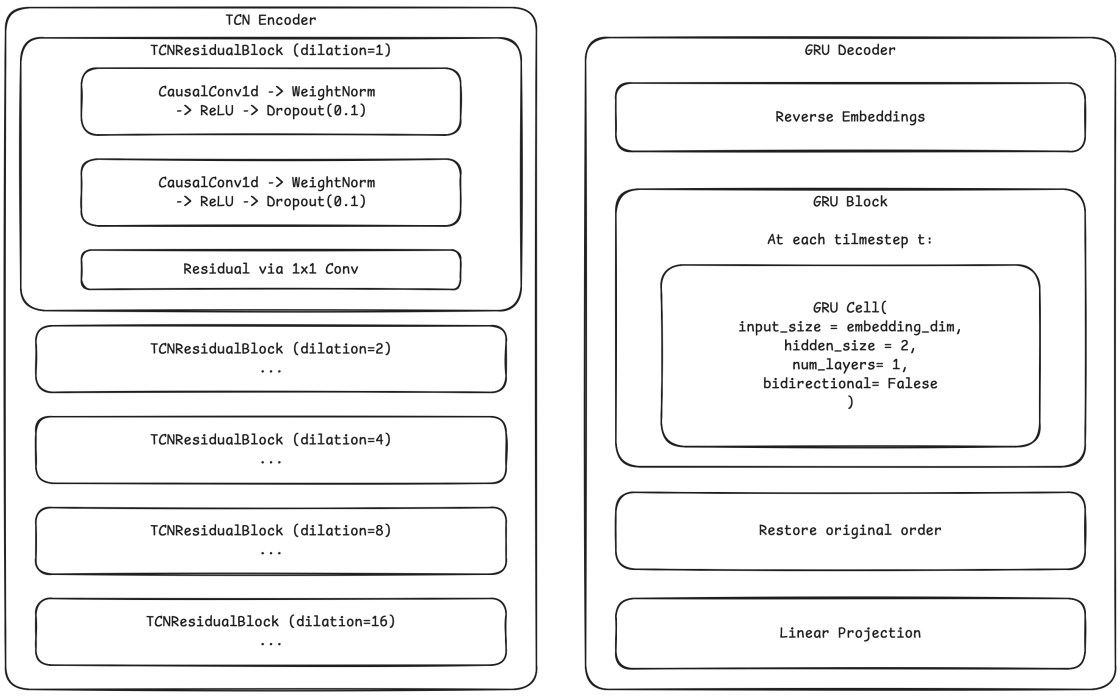


Figure 4.3: TCN-RNN Architecture

The TCN encoder consists of a stack of temporal convolutional network (TCN) residual blocks with increasing dilation factors. Each TCN block is primarily composed of two causal convolutional layers, with normalization, a nonlinear activation function (ReLU), and optional dropout for regularization. Additionally, each TCN residual block implements a residual connection to stabilize training. The output is a continuous latent representation whose size is controlled by the downsampling factor and channel width. As in Zheng and Zhang [22], there is no downsampling in the decoder; any downsampling occurs in the linear bottleneck, which is described in the next paragraph. As a result, the input shape of the encoder is equal to its output shape. The detailed TCN encoder architecture is shown in Figure 4.4a. In addition to common design choices such as dropout, residual connections, and the use of ReLU non-linearity, this encoder includes several noteworthy elements. It is designed with training efficiency and information preservation in mind. To this

end, the authors use exponential dilations and additional layers to achieve stronger long-range temporal modeling. This design intentionally trades some computational efficiency for modeling capacity. The authors also employ weight normalization to improve training stability, which further increases computational cost.



(a) TCN Encoder (b) GRU Decoder

Figure 4.4: Detailed TCN-RNN Components

After the TCN encoder, the linear bottleneck provides the only source of geometric compression within the TCN-RNN architecture. This layer is a linear projection that reduces the input channel dimension to one. In the case of Zheng and Zhang [22], two input channels were used, resulting in a compression ratio of 2:1. This means that the linear bottleneck layer of the TCN-RNN architecture produces a latent representation of the form $(\text{batch}, 1, \text{seq\_length})$.

The continuous latent representation produced by the linear bottleneck serves as the input to the decoder. The decoder has two main functions. First, the embeddings are reversed for reconstruction in reverse order, meaning that the sequence is processed from end to beginning. The authors use this strategy to help the model focus on short-range temporal correlations. Second, the decoder uses a gated recurrent unit (GRU) to reconstruct the input. GRUs are a variant of RNNs that introduce reset and update gates to control the hidden state. This GRU cell expands the latent representation compressed by the bottleneck back to its original dimensionality. Afterwards, the original order is restored and a final linear projection is applied. Figure 4.4b shows this decoder setup in more detail.

The loss function used for the TCN-RNN baseline is identical to the loss function outlined by Zheng and Zhang [22]. They use a simple MSE loss, but with the

important caveat that they use the sum-reduced variant of MSE. The following equation shows the exact loss formula used:

$$L_{\text{MSE}} = \frac{1}{T} \sum_{t=1}^{T} \sum_{d=1}^{D} \left(x_t^d - \hat{x}_t^d \right)^2 \tag{4.3}$$

This function is used as the loss function for the TCN-RNN model, while the mean-reduced version of MSE is used to evaluate all the models, as described earlier in this chapter.

4.6.1.2 Baseline 2: EdgeCodec

The EdgeCodec implementation leverages residual vector quantization (RVQ) and a heavyweight decoder to enable a lightweight encoder architecture, which is edge-optimized [24]. The compression mechanism of the EdgeCodec architecture is fundamentally different from the linear bottleneck employed by the TCN-RNN model. The compression achieved by EdgeCodec is a mixture of geometric compression in the encoder and compression in the information theoretic sense through quantization. The EdgeCodec encoder compresses both along the channel and the temporal axis and works as a dimensionality reduction mechanism. The quantization mechanism employed by EdgeCodec is a 4-layer hierarchical RVQ setup. The full EdgeCodec architecture is displayed in Figure 4.5.

Hodo et al. [24] provide the full code for the EdgeCodec implementation in a GitHub repository [49]. Therefore the EdgeCodec implementation for this thesis is largely based on this GitHub repository. An important note is that the original EdgeCodec implementation has an optional adversarial training component. For the purposes of this thesis, this is omitted, as preliminary results showed that the adversarial training component did not improve the results or was too difficult to accurately tune in the scope of this thesis.

The EdgeCodec encoder is intentionally lightweight. In essence, the EdgeCodec encoder architecture is fairly similar to the TCN-RNN encoder architecture. Like with the TCN-RNN encoder, the main components of the EdgeCodec encoder are residual units, which contain convolutional layers to capture local patterns. There are, however, notable differences. For one, TCN-RNN has five residual units with increasing dilation compared to the four residual units with steady dilations in the EdgeCodec encoder. Further, the EdgeCodec encoder does not use any weight normalization to optimize for computational efficiency. Another major difference between the EdgeCodec encoder and the TCN-RNN counterpart is the focus on maximizing the compression ability of the encoder. The TCN-RNN encoder employs no downsampling, neither temporal nor channel-wise. The EdgeCodec encoder actually does both. Through increasing strides, the EdgeCodec encoder is able to compress along the temporal axis, and through decreasing the input dimensionality, the EdgeCodec encoder compresses channel-wise. The compression ratio (CR) of the EdgeCodec encoder is configurable through these two mechanisms. The detailed EdgeCodec encoder architecture with example configurations is shown in Figure 4.6.

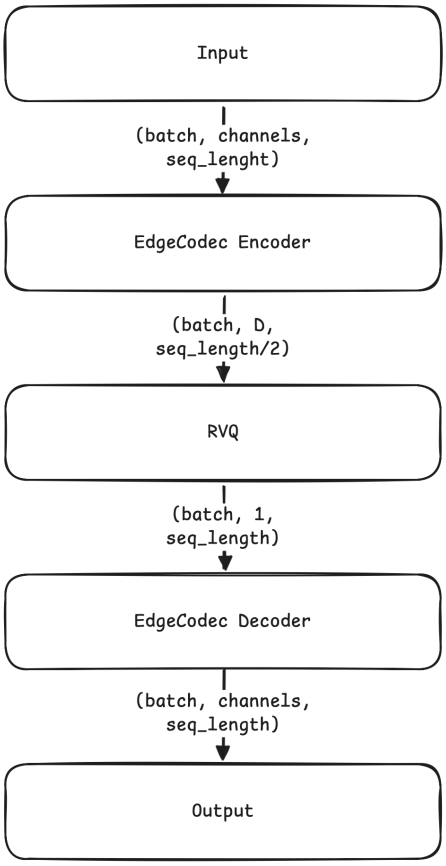


Figure 4.5: EdgeCodec Architecture

Because the EdgeCodec encoder is purposefully kept lightweight, the authors compensate through a more complex and heavyweight decoder. The EdgeCodec decoder mainly implements this through channel-wise linear layers that process each channel independently. Two of these channel-wise linear layers are implemented in the EdgeCodec decoder architecture, one at the beginning and one in the middle after the first two decoder blocks. The decoder blocks are also more heavyweight compared to the encoder blocks used in the EdgeCodec encoder. Instead of only one residual unit, the decoder block implements three residual units with increasing dilation to enhance its capabilities in modeling long-range temporal patterns. The downsampling pattern along the temporal and channel-wise axis implemented in the EdgeCodec encoder is mirrored by the EdgeCodec decoder to restore the original input shape. A detailed architectural overview of the EdgeCodec decoder is presented in Figure 4.7.

Although differences in the encoder and decoder structure between the TCN-RNN and the EdgeCodec architecture are substantial, the most substantial difference between the TCN-RNN model and the EdgeCodec model is the use of a quantization mechanism to produce discrete latents. EdgeCodec utilizes a 4-level hierarchical RVQ setup to learn the codebooks. The implementation of RVQ is standardized in most modern neural compression methods through the *vector-quantize-pytorch* library by Phil Wang [50]. Hodo et al. [24] initialize the RVQ method using embedding dimensions equal to the sequence length divided by the temporal compression of the

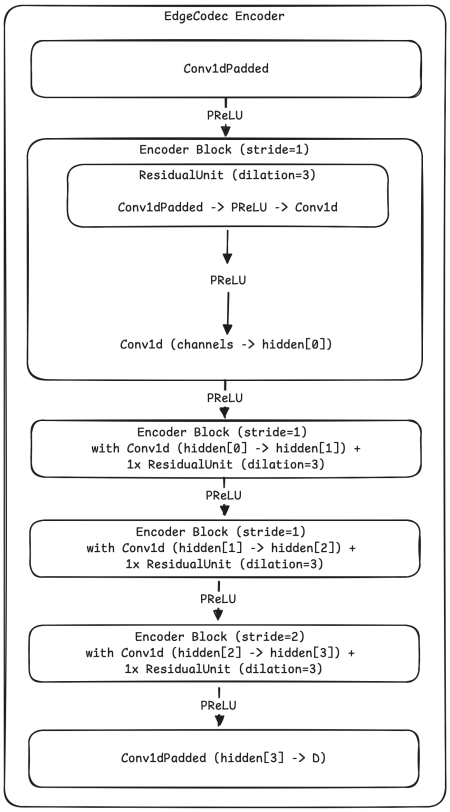


Figure 4.6: EdgeCodec Encoder

encoder. For example, if the input has a sequence length of 128, and the encoder compresses this with a temporal CR of 2:1, the embedding dimensions for the RVQ method are 64. This enables the RVQ mechanism to assign one codebook per sequence for each channel. The non-differentiability and the resulting challenging end-to-end training of quantization methods has already been mentioned in Chapter 3. RVQ handles this by employing exponential moving average (EMA) training, which enables more stable codebook updates compared to gradient-based training.

The loss function used to train EdgeCodec combines reconstruction error, quantization loss, and optional adversarial loss:

$$L(x, \hat{x}, z, \tilde{z}, y, g) = \alpha \cdot L_{\text{MSE}}(x, \hat{x}) + (1-\alpha) \cdot L_{l1-\text{smooth}}(x, \hat{x}) + \eta \cdot L_{\text{RVQ}}(z, \tilde{z}) + \gamma \cdot L_{\text{adv}}(y, g) \tag{4.4}$$

where x is the input, $\hat{x}$ is the reconstructed output, z and $\tilde{z}$ are the original and quantized latents, y and g represent the discriminator and generator outputs, and α, η, γ are weighting coefficients. As described before, for the purposes of this thesis, the adversarial training loss is omitted.

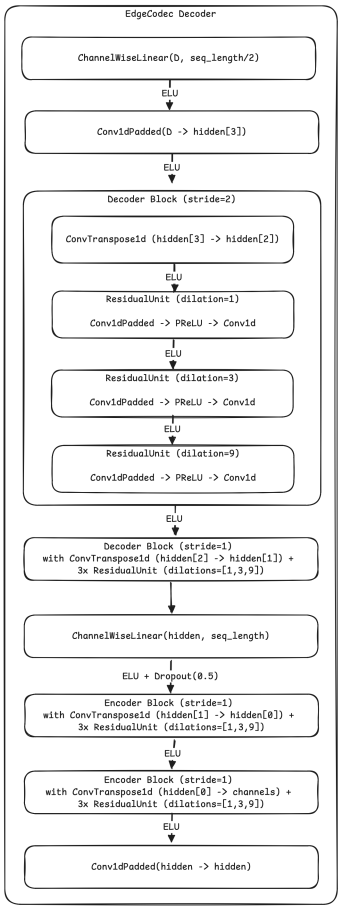


Figure 4.7: EdgeCodec Decoder

4.6.2 Proposed Implementations

It is expected that the EdgeCodec baseline implementation performs much better than the TCN-RNN baseline. This is because of two reasons. First, discrete-latent neural compression methods have been shown to consistently outperform continuous-latent neural compression methods like TCN-RNN. Second, EdgeCodec is optimized for the specific use case of compressing sensor data on the edge, which nicely fits the use case of this thesis. Given this, the proposed improvements upon the baseline models are based on the EdgeCodec architecture.

While it is expected that the EdgeCodec baseline will outperform the TCN-RNN baseline, there are a few noteworthy shortcomings of the EdgeCodec architecture. When looking at Figure 4.1 and Figure 4.2, it becomes apparent that the signals in these two datasets have completely contrasting characteristics. This is not unusual in IoT or automotive data, which is why these types of data are usually referred to as multimodal. The EdgeCodec architecture handles each signal the same. All signals share one encoder, one codebook, and one decoder. It is therefore likely that signal-specific nuances will be hard to capture for the EdgeCodec model, especially for signal types that the model is not primarily designed to handle (for example, discontinuous signals). Another potential limitation of the EdgeCodec architecture

for this thesis's use case is that the main optimization goal is reconstruction of the data, rather than downstream task utility. With the UniEdgeCodec architecture, this thesis will primarily address the first of these shortcomings, while the TAEdgeCodec tries to intentionally focus on improving downstream task utility retention.

Both UniEdgeCodec and TAEdgeCodec keep the architecture, including the encoder setup, the decoder setup, and the RVQ setup (aside from the partitioning of the codebook for UniEdgeCodec), the same compared to the EdgeCodec architecture. This is done to ensure comparability and keep the efficient architecture of the EdgeCodec model intact. The specific dimensions and parameters used for the individual experiments are introduced in the next sections, where the experiment setup will be presented.

4.6.2.1 UniEdgeCodec

At the core of the UniEdgeCodec idea is the desire to group IoT/automotive signals and handle each group with a more specialized codebook and loss function to account for group-specific characteristics more clearly. The idea to partition the codebook for different types of inputs is presented by Jiang2025 et al. [51] in their UniCodec architecture. Here, the authors try to address the problem of handling multi-domain audio data with a single codebook. Building on this idea, UniEdgeCodec implements a partitioning of the codebook by signal type. In practice, this means that each signal type will be assigned a specified range of codes in the codebook. In addition to the partitioning of the codebook, each signal type will receive a specific loss function in the UniEdgeCodec architecture. The signal types that are defined by the UniEdgeCodec architecture are continuous vs. discontinuous signals and smooth vs. irregular signals. Within the scope of this thesis, the classification of these signal types is done manually and not learned.

When looking at the signals presented in Figure 4.1 and Figure 4.2, two key characteristics, where the signals differ from one another, can be identified. One is whether a signal is continuous or discontinuous. Continuous signals like the Temp 1 signal of the Volvo EMOB dataset, or the Press_mm_hg signal of the appliances energy dataset should be easily captured by the MSE + SmoothL1 loss already implemented in the EdgeCodec architecture. Discontinuous signals like the Amps signal of the Volvo EMOB dataset or the Windspeed signal of the appliances energy dataset are likely to cause problems. The loss function of the EdgeCodec architecture heavily favors continuous signals and discourages flat regions and sharp edges. Because of this, an additional loss term is introduced for the discontinuous signals. Here, inspiration is drawn from total variation (TV) regularization. TV regularization, as introduced by Rudin et al. [52], is a popular image processing technique that is applied to preserve edges while removing noise. The assumption of TV regularization is that images are piecewise constant with sparse gradients. Optimizing image processing with TV regularization tries to minimize the total variation to preserve the piecewise constant structure of the image. This idea has seen widespread applications in the image processing domain. One example is the work presented by Liu et al. [53], who use TV regularization in combination with CNNs for image restoration. Building upon the idea of TV regularization, UniEdgeCodec applies the following reconstruction

loss function to discontinuous signals:

$$L_{recon}(x, \hat{x}, \lambda_{TV}) = L_{\text{MSE}}(x, \hat{x}) + \lambda \cdot L_{\text{TV}}(\hat{x}) \tag{4.5}$$

With $L_{\text{TV}}(\tilde{x})$ being the mean of the first order differences of the predictions.

The second characteristic to differentiate the signals presented in Figure 4.1 and Figure 4.2 is smoothness. Signals like Temp 1 in the Volvo EMOB dataset, and lights or Press_mm_hg in the appliances energy dataset are classified as smooth. These signals largely have smooth transitions and no rapid increases or decreases. Signals Rh_out or Windspeed in the appliances energy dataset, on the other hand, are classified as irregular. The transitions of these signals are rapid, leading to spiky and irregular signal behavior. When again looking at the MSE + SmoothL1 loss implemented by the EdgeCodec architecture, it is expected that these rapid changes in signal behavior are difficult to capture. It is expected that the loss implemented by EdgeCodec smooths these signals too much, leading to poorer reconstruction. To combat this, UniEdgeCodec again introduces a specific loss for irregular signals. Here, inspiration from perceptual loss methods is drawn. A perceptual loss is defined as a loss function that estimates some part of human perception with the goal of having the model predict closer to this human perception [54]. This idea is implemented in the UniEdgeCodec architecture by first identifying peaks and valleys through per-batch percentile computation, and second applying a specific MSE loss on these sections. The resulting reconstruction loss function, that is applied to irregular signals in the UniEdgeCodec architecture, is:

$$L_{recon}(x, \hat{x}, \lambda_{peak}) = L_{\text{MSE}}(x, \hat{x}) + \lambda \cdot L_{\text{peak}}(x, \hat{x}) \tag{4.6}$$

Through this grouping of the signals, four different signal types can be defined: smooth & continuous signals, smooth & discontinuous signals, irregular & continuous signals, and irregular & discontinuous signals. For smooth & continuous signals, the original MSE + SmoothL1 loss from EdgeCodec is used. For smooth & discontinuous signals, the presented TV regularization loss is used. For irregular & continuous signals, the presented perceptual loss is used. And lastly, for the irregular & discontinuous signals, a combination of the TV regularization loss and the perceptual loss is used. UniEdgeCodec combines each signal-specific loss and computes a weighted average as a combined reconstruction loss. This means that signal types need to be defined before training, so that each signal can be routed to the specific codebook partition and loss function. The complete reconstruction loss function is computed as follows:

$$\begin{aligned} L_{recon}(x, \hat{x}, \alpha, \lambda_{TV}, \lambda_{peak}, \lambda_{TV+peak}) = {} & \alpha \cdot L_{\text{MSE}}(x, \hat{x}) + (1 - \alpha) \cdot L_{l1-\text{smooth}}(x, \hat{x}) \\ & + \lambda_{TV} \cdot L_{\text{TV}}(\hat{x}) \\ & + \lambda_{peak} \cdot L_{\text{peak}}(x, \hat{x}) \\ & + \lambda_{TV+peak} \cdot L_{\text{TV+peak}}(x, \hat{x}) \end{aligned} \tag{4.7}$$

Lastly, the RVQ loss is added, just as in the EdgeCodec architecture. The total loss function for UniEdgeCodec is therefore:

$$L(x, \hat{x}, z, \tilde{z}, \lambda_{TV}, \lambda_{peak}, \lambda_{TV+peak}, \eta) = L_{recon}(x, \hat{x}, \lambda_{TV}, \lambda_{peak}, \lambda_{TV+peak}) + \eta \cdot L_{\text{RVQ}}(z, \tilde{z}) \tag{4.8}$$

4.6.2.2 TAEdgeCodec

Task-aware or task-oriented neural compression has seen an increase in popularity as more and more data is purely used for downstream task models and not human perception. As Yang et al. [18] note, here perceptual quality is irrelevant and the main focus lies on optimizing for the IB problem introduced in Chapter 3. Making a compression model task-aware is achieved by either pre-training or end-to-end training a task head and adding the task loss to the autoencoder loss [18], [55].

To add task-awareness to the EdgeCodec architecture, an end-to-end approach is implemented, which, depending on the dataset, learns a simple classification or regression head and adds the task-loss to the autoencoder loss. The resulting loss function looks as follows:

$$L(x, \hat{x}, z, \tilde{z}, y, \hat{y}) = \alpha \cdot L_{\text{MSE}}(x, \hat{x}) + (1-\alpha) \cdot L_{l1-\text{smooth}}(x, \hat{x}) + \eta \cdot L_{\text{RVQ}}(z, \tilde{z}) + \gamma \cdot L_{\text{task}}(y, \hat{y}) \tag{4.9}$$

The task loss L_{task} is defined as a smooth l1 huber loss for the regression task and a cross entropy loss for the classification task. The targets are directly loaded from the datasets, so that the reconstructed time series can be fed directly to the task head together with targets to produce task predictions and calculate the specific loss. For the regression task, the complete sequence for each window is constructed so that each timestep values is predicted by the task. The classification task tries to predict anomalies on a window-level. The task head for both tasks is identical and purposefully kept simple to isolate the compression method effect. The complete task head architecture is presented in Figure 4.8.

4.7 Training Details

In order to run the training there are certain hyperparameters which need to be decided upon for all variations of the experiments. These are parameters which will be fixed at the same value for all experiments. These parameters include the batch size and the random seed. The batch size used is allways 64 and the random seed is 42 for reproducibility. Gradient clipping is also applied with a max norm of 1.0, and mixed precision training is enabled with a precision of 32 bits for all experiments. The optimizer used for all experiments is Adam and the learning rate is set to 1e-3 for all experiments on the Emob dataset and the TCN-RNN model on the smart home dataset. All other experiments on the smart home dataset use a learning rate of 5e-4. This is done to account for the fact that the smart home dataset is smaller

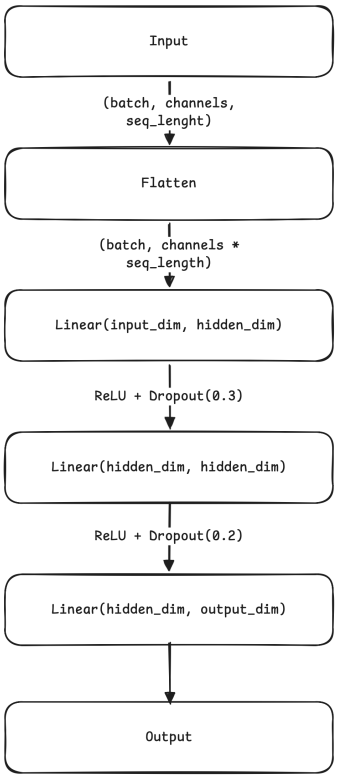


Figure 4.8: Task Head Architecture

and more complex, which makes it more difficult to train on. For all models which need to balance the MSE loss with other losses, the alpha parameter is set to 0.5, meaning that the MSE loss and the other loss are weighted equally. For all models which need to balance the reconstruction loss with the RVQ loss, the eta parameter is set to 0.25, meaning that the reconstruction loss is weighted more heavily than the RVQ loss. The specific hyperparameters for each model are introduced in more detail in the next sections.

4.7.1 TCN-RNN

For the baseline model, the relevant hyperparameters were set based on the implementation in the original paper by Zheng and Zhang [22]. The TCN encoder is implemented with 5 residual blocks, with a channel progression of [3, 3, 3, 3, 3] for the Emob dataset and [7, 7, 7, 7, 7] for the smart home dataset. The kernel size is set to 4 and the dropout is set to 0.1. The linear bottleneck compresses the channel dimension from the orignal number of channels to 1, resulting in a compression ratio of x:1 where x is the initial number of channels. The GRU decoder is implemented with a hidden size of 3 and 7 respectively across 1 layer. The loss function used for training is the sum-reduced MSE loss, as described in the methods section. The decoder is not bidirectional, and the sequence is reversed before being fed to the decoder. The maximum number of epochs is set to 500, with an early stopping mechanism that stops training if the validation loss does not improve for 10 consecutive epochs with a minimum improvement threshold of 1e-4 and the l2 regularization strength is set

to 0. The learning rate scheduler used is cosine annealing with 500 epochs and a minimum learning rate of 1e-6. The optimizer uses betas of (0.9, 0.999) and epsilon of 1e-08. The pytorch lightning trainer is configured with the same maximum epochs and early stopping mechanism as described before.

4.7.2 EdgeCodec

For the EdgeCodec baseline, the hyperparameters were initially set based on the original implementation by Hodo et al. [24]. However, the paper is not fully clear on the exact hyperparameters used for training and the publicly available codebase does not provide exact details about the training. Therefore, while the original implementation served as a starting point, the hyperparameters were adjusted where it was considered necessary or beneficial to fit the data. The codebook size and betas are the two parameters which are directly taken from the original implementation, all other parameters have been adjusted.

The channel progression in the encoder is [7, 7, 7, 7] for the smart home dataset and [3, 3, 3, 3] for the Emob dataset, matching the number of initial channels. The quantizer has 2 stages of residual quantization with a codebook size of 768 per quantizer and an embedding dimension of 32 for the Emob dataset and 72 for the smart home dataset. This is done in order to the embedding dimension to be equal to the sequence length divided by the temporal compression ratio of the encoder, as described in the methods section. The decoder then uses a identical channel progression as the encoder. The EdgeCodec uses an optimizer with betas of (0.9, 0.98) and epsilon of 1e-8. The pytorch lightning trainer is configured with maximum epochs of 80. The early stopping mechanism is configured with a patience of 20 epochs and a minimum delta of 1e-2. The scheduler used is a cosine annealing scheduler with a maximum of 500 and a minimum learning rate of 1e-6.

4.7.3 Uni-EdgeCodec

The UniEdgeCodec remains largely similar to the EdgeCodec in terms of architecture and hyperparameter settings. The values used for the encoder, decoder, optimizer and the pytorch lightning trainer are the same as those used for the EdgeCodec. The quantizer uses a codebook with the size of 1000 partitioned into smooth-continuous [0 - 300], smooth-discontinuous [300 - 300], irregular-continuous [300 - 600] and irregular-discontinuous [600 - 1000] partitions. The signals are categorized as follows: lights and HR_out are categorized as smooth-continuous, Press_mg_hg and T_out are categorized as irregular-continuous and Windspeed, Visibility and Tdewpoint are categorized as irregular-discontinuous. The embedding dimensions and commitment cost of the RVQ varies between datasets as the Emob dataset has a embedding dimension of 64 and a commitment cost of 0.75 while the smart home dataset has an embedding dimension of 72, similar to the EdgeCodec implementation, and a commitment cost of 0.35. The scheduler used is a cosine annealing scheduler with a maximum of 80 and a minimum learning rate of 1e-6. Additionally, the Uni-EdgeCodec uses a lambda of 0.1 for the tv loss, a peak weight of 1.0 for the peak loss, a peak percentile of 90, a discontinuous weight of 10.0 for the TV loss, a perceptual

weight of 1.0 for the peak loss and a discontinuous perceptual weight of 1.0 for the combined TV and peak loss.

4.7.4 Task Aware EdgeCodec

The task aware EdgeCodec also share many hyperparameters with the base EdgeCodec model. These include the parameters for the encoder, decoder, optimizer, scheduler and the pytorch lightning trainer. The quantizer also has a codebook with the size of 768 and an embedding dimension of 64. The main difference is the training module which now has a beta for balancing the task loss. This value is set to 1e-2 and the dimension for the task head is set to 128.

4.7.5 Compression Ratio

Certain parameters have been set with the intent of roughly alligning the CR of each codec so that their results will be comparable. For this reason, the codebook size and the number of quantizers have been modified. The CR can be computed as follows:

$$\text{original\_size} = \text{channels} \times \text{timesteps} \times 32 \tag{4.10}$$

$$\text{compressed\_size} = \lceil \log_2(\text{codebook\_size}) \rceil \times \text{channels} \times \text{num\_quantizers} \tag{4.11}$$

$$\text{CR} = \frac{\text{original\_size}}{\text{compressed\_size}} \tag{4.12}$$

Table 4.1 presents the computed compression ratios for the three architectures evaluated on the Volvo EMOB dataset, demonstrating that the architectures achieve comparable compression performance despite their different internal configurations.

Table 4.1: Compression Ratio Comparison for EMOB Dataset (3 channels, 128 timesteps)

Model	**Codebook Size**	**Emb. Dim.**	**Num Quant.**	**CR**
EdgeCodec	768	32	2	213.7×
UniEdgeCodec	1000	64	2	204.8×
TAEdgeCodec	768	64	2	213.7×

Similarly, Table 4.2 shows the computed compression ratios for the appliances energy dataset. Despite the larger number of channels and longer window size compared to the EMOB dataset, all three architectures maintain comparable compression performance, achieving between 240× and 230× compression.

Table 4.2: Compression Ratio Comparison for Appliances Energy Dataset (7 channels, 144 timesteps)

Model	**Codebook Size**	**Emb. Dim.**	**Num Quant.**	**CR**
EdgeCodec	768	72	2	240.4×
UniEdgeCodec	1000	72	2	230.4×
TAEdgeCodec	768	64	2	240.4×

5

Experimental Results

Within this chapter the results for the experiments for phase one and two are presented. The main goal is to answer research questions one and two.

5.1 Phase 1 Experiments

First, the efficacy of the two baseline models presented in Chapter 4 when applied to IoT and automotive data is determined. For this, both these baseline models were implemented according to the architecture outlined in Chapter 4.

5.1.1 The Problem with Continuous-latent Autoencoders

The TCN-RNN continuous-latent autoencoder is first evaluated on the appliances energy dataset used by Zheng and Zhang [22], [48]. The use case the authors present in their paper uses two channels: "Appliances" and "Windspeed". The training goal is to compress and reconstruct these two signals. The training setup for this first experiment is purposefully kept as close as possible to the authors setup.

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5.1.2 Applying EdgeCodec to Multimodal IoT and Automotive Data

Because the EdgeCodec architecture was copied almost one to one from the EdgeCodec Github repository, the implementation correctness will not be tested on a deticated use case. Instead, EdgeCodec is directly applied to

6

Conclusion

You may consider to instead divide this chapter into discussion of the results and a summary.

6.1 Discussion

6.2 Conclusion

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Appendix 1